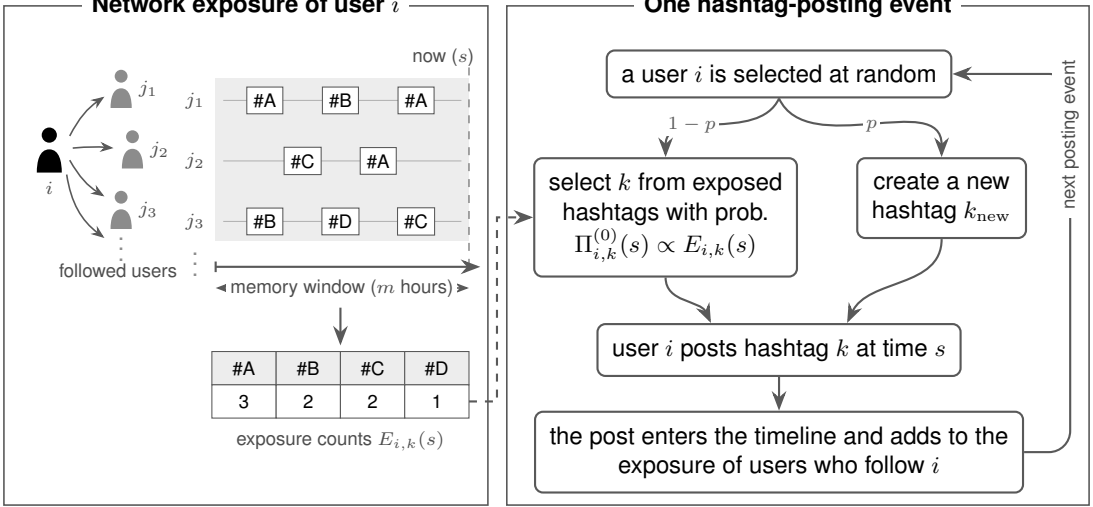

Network exposure of user i
now (s)
j_1
j_2
j_3
i
#A
#B
#A
#C
#A
#B
#D
#C
followed users
memory window (m hours)
#A
#B
#C
#D
3
2
2
1
exposure counts $E_{i,k}(s)$
One hashtag-posting event
a user i is selected at random
$1-p$
p
select k from exposed hashtags with prob. $\Pi^{(0)}_{i,k}(s) \propto E_{i,k}(s)$
create a new hashtag k_{new}
user i posts hashtag k at time s
the post enters the timeline and adds to the exposure of users who follow i
next posting event